
Displacement, Velocity and Acceleration Graphs Review

A complete lesson-linked revision guide for Lampros Andrinopoulos's 15 July 2026 lesson

Core skill

Translate between displacement-time, velocity-time and acceleration-time graphs using gradients, signed areas and physical meaning.

Lesson questions

Q001 to Q011, ordered from definitions to graph conversion and bouncing motion.

Specification match

Issue 3, Unit 1, section 1.3, points 2 and 3; terminal velocity also appears in point 9.

Evidence boundary

Question-specific status, source certainty and student/tutor distinctions are retained.

How to read this guide. "Your attempt" records what the learner first said or did. "Tutor correction" records the lesson's guided method. "Secure answer" identifies an answer verified from lesson and source-controlled question evidence; it does not imply that an official mark scheme is present. Boxes marked **Derived expansion** add general physics teaching and are not presented as quotations or as reproductions of the tutor's board.

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Best use: read Sections 1-3 once, then cover the answers and redo Q004-Q011 aloud. A correct option without a gradient/area/sign explanation is not secure understanding.

1. What Edexcel expects

The specification requires you to draw and interpret displacement-time, velocity-time and acceleration-time graphs, and to obtain physical quantities from their slopes and areas, including non-uniform acceleration. The lesson directly trained those skills.

Interpret

- State what the sign and gradient mean physically.
- Distinguish velocity from speed and displacement from distance.
- Use the initial condition: an area gives a *change*, not necessarily the quantity itself.

Convert

- Convert one graph segment at a time.
- Match gradients at boundaries where the motion is continuous.
- Show turning points where velocity is zero.
- State idealisations at discontinuities.

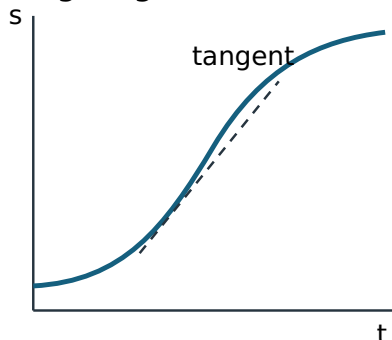
Exam habit: read both axes and units before touching the graph. Underline decisive modelling words such as *from rest*, *falls freely*, *constant velocity* and *uniform acceleration*.

2. The three-graph dictionary

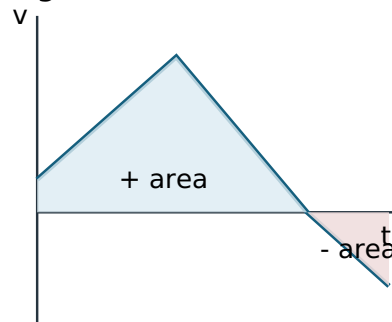
Graph	Gradient tells you	Area tells you	High-value interpretation
Displacement-time, s-t	Velocity, $v = ds/dt$	No standard motion quantity at this level	Horizontal means at rest. Positive/negative slope means positive/negative velocity. Steeper means larger speed.
Velocity-time, v-t	Acceleration, $a = dv/dt$	Signed displacement, $\Delta s = \int v \, dt$	Area below the time axis is negative displacement. Distance requires absolute area.
Acceleration-time, a-t	Rate of change of acceleration (jerk), not normally required here	Change in velocity, $\Delta v = \int a \, dt$	To recover velocity, add the initial velocity: $v(t) = u + \int a \, dt$.

Average velocity = total displacement / total time | **Average speed** = total distance / total time

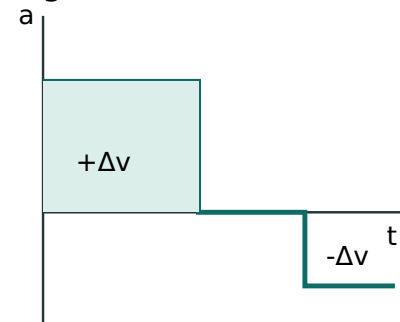
s-t graph
tangent gradient = v



v-t graph
signed area = Δs



a-t graph
signed area = Δv



Derived overview. Always decide whether an area is signed (displacement/change in velocity) or absolute (distance).

3. Signs, units and areas

Signs

Velocity says which way the object moves. Acceleration says which way velocity changes. The object speeds up when velocity and acceleration have the same sign, and slows when their signs are opposite.

Units

s in m; t in s; v in m s^{-1} ; a in m s^{-2} . Gradient units divide vertical by horizontal. Area units multiply.

Zero is contextual

$v = 0$ can mean release, a turning point, or a zero crossing during contact. $a = 0$ can mean constant non-zero velocity, rest, or terminal velocity.

Operation	Unit check	Result
Gradient of s-t	m / s	m s^{-1} (velocity)
Gradient of v-t	$(\text{m s}^{-1}) / \text{s}$	m s^{-2} (acceleration)
Area under v-t	$(\text{m s}^{-1})(\text{s})$	m (displacement)
Area under a-t	$(\text{m s}^{-2})(\text{s})$	m s^{-1} (change in velocity)

Distance from a v-t graph: calculate the magnitude of every positive and negative area, then add. Net displacement instead adds signed areas, so cancellation is allowed.

4. Q001-Q003: foundations

Q001 - RECORDING-DERIVED CONCEPT CHECK

SOLVED

When is distance equal to the magnitude of displacement?

Your attempt: "When travelling in a straight line." This captures the geometry but not the direction condition.

Tutor correction: compare the direct path with a longer curved path, then use the next out-and-back example to reveal the missing condition.

Secure answer: distance equals the magnitude of displacement only when the object travels directly along a straight line from start to finish without reversing direction.

Evidence: lesson question Q001. This is a tutor concept check, not an official exam item.

Q002 - RECORDING-DERIVED CONCEPT CHECK

SOLVED

Move away, then return to the start

Your attempt: displacement is zero.

Tutor confirmation: the final position equals the initial position, so the displacement vector is zero; the distance is twice the outward path length.

Answer: displacement = 0, but distance > 0.

Trap: displacement is not "how much ground was covered". It depends only on initial and final position.

Q003 - ACCELERATION-TIME INTERPRETATION

COMPLETED WITH
PROMPTING

What can an a-t graph tell you when the initial velocity changes?

Given: acceleration begins positive, decreases through zero and later becomes negative. Compare $u = 0$ with $u = -5 \text{ m s}^{-1}$. No numerical scale was certified.

Your attempt: you asked for repetition and later reasoned that, in the tutor's illustrated negative-velocity case, the object could slow, stop and turn right.

Tutor correction: area under a-t gives *change in velocity*, not velocity itself. Positive acceleration while v is negative makes the velocity less negative, so the object slows. Reversal occurs only when the accumulated positive area is large enough to bring v to zero and then positive.

$$v(t) = u + \int_0^t a(\tau) d\tau$$

Condition for reversal from $u = -5 \text{ m s}^{-1}$: the cumulative signed acceleration area must reach $+5 \text{ m s}^{-1}$ to reach $v = 0$, and exceed it to become positive. Positive acceleration alone is not enough.

Sign trap: negative acceleration does not automatically mean slowing down. If v is also negative, the object speeds up in the negative direction.

Evidence limitation: the drawn curve had no certified scale, so this guide states the reversal condition rather than claiming that every such curve must cause reversal.

5. Q004-Q006: core exam graph skills

Q004 - EDEXCEL 6PH01/01, JUNE 2009 SERIES, Q8 - 1 MARK

SOLVED

RECOVERED SOURCE

Total distance in 4 s

Given graph: velocity rises linearly from 0 to 20 m s^{-1} during 0-2 s, then stays at 20 m s^{-1} until 4 s. Options: 20 m, 40 m, 60 m, 80 m.

Your attempt: correctly chose area under the graph and split it into a triangle plus rectangle.

Tutor method: triangle = $\frac{1}{2} \times 2 \times 20 = 20 \text{ m}$; rectangle = $2 \times 20 = 40 \text{ m}$. Alternative: treat the whole region as a trapezium.

$$\text{distance} = 20 \text{ m} + 40 \text{ m} = 60 \text{ m}$$

Secure answer: C, 60 m.

Checks: velocity never goes below zero, so signed area equals distance. Units: $(\text{m s}^{-1})(\text{s}) = \text{m}$.

Identifier recovered from the exact wording/values: Edexcel GCE Physics 6PH01/01, June 2009 series (exam date 21 May 2009), Q8; PMT Motion Graphs MCQ compilation Q11.

Q005 - OCR H157/01, 24 MAY 2016, Q10 - 1 MARK

PROMPTED

RECOVERED SOURCE

Bouncing golf ball: first maximum height after the bounce

Displayed sign convention: falling velocity is positive; rising velocity is negative.

Your attempt: selected C because $v = 0$ at maximum height, then correctly recognised that the choice had been partly lucky because another zero crossing existed.

Tutor reconstruction: release at $v = 0$; falling speed increases; during floor contact the ball deforms and velocity crosses zero at A; it leaves the floor moving upward near B; gravity reduces the upward speed; the first post-bounce apex occurs at C, where $v = 0$.

Secure answer: C. A is a zero-velocity instant during contact, not the requested post-bounce apex.

Zero-crossing method: do not choose " $v = 0$ " blindly. Name the physical phase immediately before and after the point.

Ideal versus real contact: a sloped transition shows finite contact time. A vertical jump is an ideal instantaneous reversal and implies an impulse-like acceleration, not an ordinary finite value.

Maximum velocity on a displacement-time graph

Your attempt: initially used velocity = displacement/time, then accepted that instantaneous velocity requires the tangent gradient and chose the steepest positive tangent.

Tutor method: draw or imagine a tangent at A, B, C and D; compare slopes. A steep negative tangent is a large negative velocity, not the maximum positive velocity.

Secure answer: B, the steepest positive tangent.

Why s/t failed: s/t from the origin gives an average over that interval. It equals instantaneous velocity only in special constant-velocity situations represented by an appropriate straight line.

The transcript's apparent option "P" is an ASR error; the official graph contains A-D and the recovered answer is B.

6. Q007-Q009: drag, terminal velocity and sensors

Object approaching terminal velocity

Given: an object is dropped from rest; its v-t curve rises from zero and approaches a constant by about 30 s. Which statement is correct?

Your attempt: you considered constant acceleration, then mixed up zero resultant force with zero drag. In the vacuum extension you briefly suggested the velocity might remain zero.

Tutor correction: the decreasing gradient means acceleration decreases. At release, $v = 0$ so drag is zero and the initial acceleration is approximately g . As speed rises, drag rises; at terminal velocity, drag equals weight, resultant force is zero, acceleration is zero, and velocity is constant but non-zero.

At terminal velocity: $\Sigma F = 0 \Rightarrow \text{drag} = \text{weight} \text{ and } a = 0$

Answer: C, acceleration is 9.81 m s^{-2} at $t = 0$.

Vacuum extension: near Earth with no air resistance, weight still acts. Acceleration remains constant at g and the v-t graph is a straight line (for a fixed sign convention).

Source status: exact content matched to an OCR A topic compilation, "Motion with non-uniform acceleration", item 8. The original paper code/session was not proven from the cropped lesson slide.

Fairground-car accelerometer

Question idea: a water-ring accelerometer is attached to a car. At Y, the displacement-time graph has an approximately straight negative slope. Choose the water surface.

Your attempt: first chose D because the velocity at Y was negative.

Tutor correction: the instrument responds to acceleration, not velocity. At Y, the s-t gradient is negative but approximately constant, so velocity is constant negative and acceleration is zero.

Answer: A, level water surface.

Two-step sensor rule: first obtain velocity from the s-t gradient; then ask whether that velocity is changing. Negative velocity by itself does not mean non-zero acceleration.

Q009 - PARACHUTIST ACCELERATION GRAPH

PROMPTED

OFFICIAL IDENTITY UNRESOLVED

Free fall, parachute opening and a new terminal speed

Your attempt: initially eliminated options inconsistently; with prompts you recognised initial constant gravitational acceleration and an acceleration opposite the downward motion when the parachute opens.

Tutor method: underline "falls freely". With downward positive, initial acceleration is $+g$ and constant. Opening the parachute creates a large upward force, so acceleration becomes negative. As downward speed falls and a new lower terminal speed is approached, acceleration tends back toward zero.

Lesson answer: B.

Enrichment: if ordinary drag before opening is included, the initial acceleration would decrease smoothly from g rather than stay exactly constant. That is a model extension, not the assumption in the printed "falls freely" interval.

Source uncertainty: a matching public specimen/pre-publication mirror was found, but the displayed slide had no board header, paper code, date, question number or visible mark. Do not cite a precise official sitting.

7. Q010: piecewise graph conversion

Q010 - TUTOR-AUTHORED/COMPILED DECK, QUESTION 1

PROMPTED

MARKS/SOURCE NOT SHOWN

Convert a piecewise displacement-time graph to v-t and a-t

Visible prompt: describe the motion, then sketch the velocity-time and acceleration-time graphs. The given s-t graph has a straight positive-slope segment, a horizontal plateau, then a downward parabolic segment crossing $s = 0$.

Your attempt: correctly identified positive motion and rest, but initially thought a curved s-t section could not produce a straight v-t section. You later mixed "constant velocity" with "velocity increasing at a constant rate".

Tutor method: translate segment by segment. Straight positive s-t means constant positive v. Flat s-t means $v = 0$. For the final parabolic interval, v starts at 0 and then decreases linearly into negative values; a is constant negative.

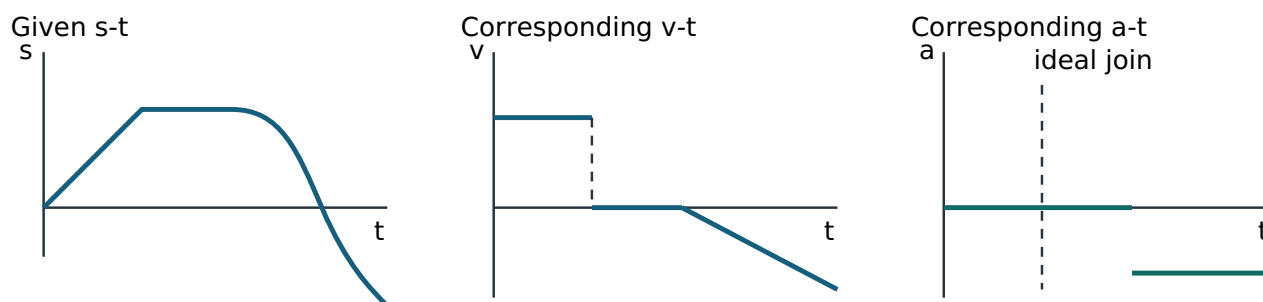
s-t segment	v-t result	a-t result within the segment	Physical description
Straight rising	Horizontal above zero	Zero	Constant positive velocity

s-t segment	v-t result	a-t result within the segment	Physical description
Horizontal plateau	Zero	Zero	At rest
Concave-down parabola	Starts at $v=0$, then becomes increasingly negative	Constant negative	Motion in the negative direction with increasing speed

Final segment-wise sketch: v-t is positive constant, then zero, then a descending straight line. a-t is zero on the open first two intervals, then constant negative on the final interval.

Derived expansion - discontinuity caveat: the ideal sharp join from a positive s-t slope to a flat plateau makes velocity jump instantly to zero. At that exact instant, ordinary acceleration is not finite; an exact model needs an impulse/undefined-at-the-instant caveat. The horizontal $a = 0$ levels describe the intervals between joins.

Derived schematic - not to scale



Straight s-t $\rightarrow$ constant v. Flat s-t $\rightarrow v = 0$. Quadratic s-t $\rightarrow$ linear v $\rightarrow$ constant a.
Dashed join = ideal discontinuity; no ordinary finite acceleration is defined at that instant.

Derived expansion from the visible pre-boundary prompt and the segment-wise physics. It is not a reproduction of the tutor's final board.

8. Q011: bouncing-ball graph conversion

Q011 - SAME TUTOR DECK, QUESTION 2

PROMPTED

QUALIFIED EVIDENCE

Convert a bouncing-ball v-t graph to s-t and a-t

Visible pre-boundary prompt: the v-t graph consists of repeated straight downward-sloping flight segments separated by ideal instantaneous upward jumps; the amplitudes decrease. Sketch s-t and a-t.

Your attempt: chose a positive release height and the floor as zero. You initially proposed a straight s-t descent, then chose a parabola by intuition but needed prompting to justify its initial horizontal tangent and the lower rebound peaks.

Tutor method: choose floor $s = 0$. Since $v(0) = 0$, the s - t graph begins with a horizontal tangent. Constant gravitational acceleration makes each free-flight s - t arc parabolic. Areas under v - t determine height changes, so diminishing areas produce lower rebound apexes. Parallel v - t flight slopes mean the same constant gravitational acceleration between impacts.

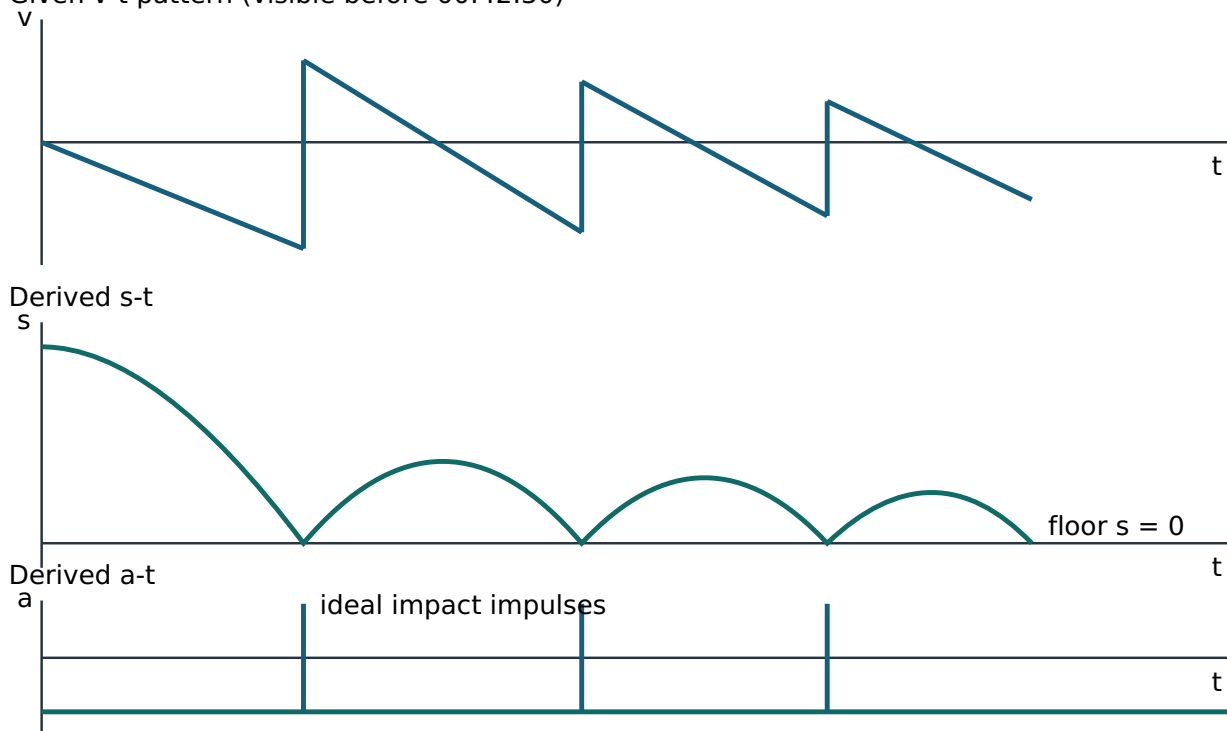
$$\text{slope of } s\text{-}t = v \quad | \quad \text{area under } v\text{-}t = \Delta s \quad | \quad \text{slope of } v\text{-}t = a$$

Secure result: s - t has repeated parabolic down/up arcs touching the floor, with successively lower apexes. a - t is constant gravitational acceleration between impacts; ideal instantaneous velocity jumps correspond to impulse-like acceleration spikes.

Idealisation: the tutor described the instantaneous-bounce spike as "plus infinity" from $a = \Delta v / \Delta t$ as Δt tends to zero. In a real bounce, contact time is finite and acceleration is very large but finite. In an ideal graph, the jump is represented by an impulse/distribution, not an ordinary measurable infinite acceleration.

Derived expansion - qualitative only, not to scale

Given v - t pattern (visible before 00:42:30)



Evidence boundary: the given sawtooth pattern and the start of the s - t construction were visible before media failure. The completed teaching above is a clearly labelled physics-derived expansion supported by the lesson transcript and the local textbook's bouncing-motion treatment; it does not claim to reproduce any post-00:42:30 board state.

Source status: exact prompt recovered from the tutor deck; no marks or external identifier were shown. Only pre-00:42:30 visuals are described. Later board geometry is not claimed.

9. Parabolas and graph conversion

Derived expansion: this section generalises the lesson's conversion method. It is supporting teaching, not a claim about a particular hidden slide.

Why constant acceleration gives a parabola in s-t

$$s = s_0 + ut + \frac{1}{2}at^2 \Rightarrow v = u + at \Rightarrow a = \text{constant}$$

- The t^2 term makes displacement quadratic in time, so the s-t graph is parabolic.
- Differentiating once gives linear v-t.
- Differentiating again gives horizontal a-t.
- The initial s-t tangent is u. If $u = 0$, the parabola starts with a horizontal tangent.

Appearance trap: do not call every smooth curved s-t line a parabola. You may infer quadratic behaviour when constant acceleration is stated, established by the v-t gradient, or otherwise justified.

Reliable segment-by-segment conversion algorithm

1. **Write the sign convention.** For vertical motion, decide whether upward or downward is positive.
2. **Mark breakpoints.** Treat each straight/curved/flat interval separately.
3. **Read slope first.** s-t slope gives v; v-t slope gives a.
4. **Read area second.** v-t area sets displacement change; a-t area sets velocity change.
5. **Use boundary conditions.** $v = 0$ means a horizontal s-t tangent; the graph itself should remain continuous unless position genuinely jumps, which physical motion normally forbids.
6. **Check signs.** Below-axis v means displacement decreases; it does not make distance decrease.
7. **Flag ideal jumps.** A vertical v-t jump means an impulsive acceleration, not a normal finite a-t level.
8. **Check dimensions and physical story.** State moving direction, speeding/slowing, rest, contact or terminal motion.

10. Worked examples

Derived examples: these are new practice expansions using the lesson rules. They are not additional recovered lesson questions.

Example A - signed area versus distance

A v-t graph is $+4 \text{ m s}^{-1}$ for 3 s, then -2 m s^{-1} for 5 s.

1. Positive displacement = $4 \times 3 = +12 \text{ m}$.
2. Negative displacement = $(-2) \times 5 = -10 \text{ m}$.
3. Net displacement = $+12 - 10 = +2 \text{ m}$.
4. Total distance = $12 + 10 = 22 \text{ m}$.

Check: displacement may cancel; distance cannot.

Example B - acceleration area and reversal

An object starts with $u = -6 \text{ m s}^{-1}$. It experiences $a = +2 \text{ m s}^{-2}$ for 4 s.

$$\Delta v = \text{area} = at = (+2)(4) = +8 \text{ m s}^{-1}; \quad v = u + \Delta v = -6 + 8 = +2 \text{ m s}^{-1}$$

It reaches $v = 0$ after 3 s, then reverses. The positive acceleration did not instantly mean positive motion; it first reduced the negative velocity.

Example C - tangent and curvature

An s-t graph is increasing but becoming less steep.

- v is positive because the slope is positive.
- Speed is decreasing because the slope magnitude decreases.
- a is negative because velocity is decreasing.

Example D - terminal velocity statement

At terminal velocity, write: "The upward resistive force equals the downward weight, so resultant force and acceleration are zero. The object continues at constant non-zero velocity." Do **not** write "all forces are zero".

11. Formula and trap sheet

Rule	Correct use	Typical trap
$v = ds/dt$	Tangent gradient of s-t	Using s/t for changing motion
$a = dv/dt$	Tangent gradient of v-t	Reading acceleration from the sign of velocity
$\Delta s = \int v \, dt$	Signed area under v-t	Calling signed area distance when v changes sign
distance = $\int v \, dt$	Add absolute areas	Allowing cumulative distance to decrease
$\Delta v = \int a \, dt$	Signed area under a-t	Forgetting to add u
speeding up	v and a have the same sign	Equating positive a with speeding up
turning point in s-t	$v = 0$; horizontal tangent	Assuming every $v = 0$ has the same physical meaning
terminal velocity	$\Sigma F = 0$, $a = 0$, v constant	Saying drag = 0 or $v = 0$
uniform acceleration	s-t quadratic, v-t linear, a-t horizontal	Inferring a parabola from appearance alone
instantaneous v jump	Ideal impulse; real contact is finite	Drawing a normal finite a value across zero time

Graph checklist

- Axes and units?
- Initial value?
- Gradient sign/magnitude?
- Area signed or absolute?
- Continuous curve or ideal jump?
- Physical story?

Language checklist

- "Constant velocity" = horizontal v-t.
- "Velocity increases at a constant rate" = straight sloping v-t.
- "At rest" means $v = 0$, not necessarily $s = 0$.
- "Zero net force" does not mean zero individual forces.

12. Focused practice and redo plan

This order targets the exact difficulties evidenced in the lesson while preserving questions already secure.

Prio rity	Re do	What to produce without notes	Pass condition
1	Q0 03	Explain $u = 0$ and $u = -5$ cases using $v(t) = u + at$.	State the $+5 \text{ m s}^{-1}$ reversal threshold and avoid "positive a means moving right".
2	Q0 06	Draw tangent comparisons at A-D.	Use instantaneous gradient, not s/t ; identify B and explain the negative-slope distractor.
3	Q0 07	Write forces at release, during acceleration and at terminal speed.	At terminal speed: drag = weight; $\Sigma F = 0$; $a = 0$; $v \neq 0$.
4	Q0 08	Use the $s-t$ slope at Y, then infer acceleration.	Say constant negative velocity means zero acceleration; answer A.
5	Q0 09	Sketch the sign sequence of acceleration with downward positive.	$+g$ during stated free fall, large negative on opening, then approach 0; answer B.
6	Q0 10	Convert each $s-t$ segment to $v-t$ and $a-t$.	Correct horizontal/zero/sloping levels plus an ideal-join caveat.
7	Q0 11	Sketch derived $s-t$ and $a-t$ from the visible sawtooth $v-t$.	Horizontal initial $s-t$ tangent, lower parabolic apexes, constant flight acceleration, impact impulses qualified.

Three-pass routine

1. **Pass 1 - explain:** speak the physical story before calculating.
2. **Pass 2 - draw:** reproduce the graph shapes and label signs, zero crossings and units.
3. **Pass 3 - exam-speed:** answer each MCQ in 60-90 seconds, then spend 30 seconds stating why the strongest distractor is wrong.

Maintenance questions: Q001, Q002 and Q004 were secure. Recheck them briefly after the higher-priority conversions rather than spending most practice time there.

13. Homework status

Status preserved from the lesson: further motion-graph practice was assigned in principle, but the exact URL, filename, question numbers, count and deadline were not captured. The tutor also promised background/link material and more motion-graph questions. Delivery of those materials is not evidenced in the controlling lesson record.

- **Confirmed action:** attempt additional motion-graph questions and bring/send the work for discussion if possible.
- **Unresolved:** the phrase "these questions" cannot safely be converted into a numbered set.
- **Next topic mentioned:** a few SUVAT questions were promised for the next rectilinear-motion stage; exact items were not captured here.
- **Not included:** no Q012 prompt, diagram, scale, exact sign sequence or solution is reconstructed in this guide.

Practical substitute until the tutor's set is recovered: use the seven-item redo plan in Section 12. This is a derived practice recommendation, not a claim that these were the tutor's exact assigned questions.

14. Sources and evidence notes

Controlling curriculum and textbook sources

1. **Pearson Edexcel International Advanced Subsidiary/Advanced Level Physics Specification, Issue 3 (July 2021).** Unit 1, section 1.3 Mechanics, points 2-3: draw/interpret s-t, v-t and a-t graphs and derive quantities from slopes/areas, including non-uniform acceleration. Point 9 expects terminal velocity. This is the canonical specification for the guide.
2. **Edexcel Physics Student Book 1, Chapter 3.3, printed pp. 36-40.** Supports tangent gradient on s-t, gradient/area on v-t, and the common-timescale bouncing-ball treatment. It is a reference expansion source, not proven as the origin of the tutor's displayed questions.
3. **Lesson evidence, 15 July 2026, Q001-Q011.** Used for chronology, student attempt, tutor correction, completion status and source uncertainty. The recording's usable visual evidence ends before 00:42:30; no later board state is described as seen.

Recovered question identifiers

- **Q004:** Edexcel GCE Physics 6PH01/01, June 2009 series (exam date 21 May 2009), Q8; also PMT Motion Graphs MCQ compilation Q11. Exact wording/values recovered; 1 mark.
- **Q005:** OCR AS Level Physics B (Advancing Physics) H157/01, 24 May 2016, Q10. First-party OCR question identity recovered; 1 mark.
- **Q006:** OCR AS Level Physics A H156/01, 15 May 2018, Q1. First-party OCR question identity recovered; 1 mark.
- **Q007:** OCR A topic compilation, "Motion with non-uniform acceleration", item 8. Exact content secure; original sitting unresolved; 1 mark shown in compilation.
- **Q008:** Cambridge Pre-U Physics 9792/01, May/June 2012, Q7. Accessible mirror matched; 1 mark.
- **Q009:** matching public specimen/pre-publication candidate only. Board/session/question number and mark remain unresolved.
- **Q010-Q011:** tutor-authored/compiled deck, "Displacement-time, velocity-time and acceleration-time graphs", Questions 1-2. No external identifier or marks shown.

Direct links where recovered

- [OCR H157/01 question paper \(Q005\)](#)
- [OCR H156/01 question paper \(Q006\)](#)
- [Edexcel 6PH01/01 original-paper mirror \(Q004\)](#)
- [PMT Edexcel Motion Graphs compilation \(Q004\)](#)
- [OCR A topic compilation mirror \(Q007\)](#)
- [Cambridge Pre-U 9792/01 mirror \(Q008\)](#)

Evidence boundary: Q012 is deliberately absent as a reconstructed question. Only Q001-Q011 are taught here. The Q011 completed solution is labelled as a derived expansion and does not claim a post-00:42:30 visual review. Homework scope remains unresolved rather than being guessed.

Prepared as a private learner-facing study guide. No raw transcript, internal filesystem paths, model logs or production metadata are included.